
Top 5 Papers for Next-Level Research Direction (draft)

1 message

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Hey Dan,

I'm planning to share this with the group as a summary of where I see us going next. Let me know if I'm missing anything important.

Hi everyone,

Following up on our recent conversations, I pulled together a short list of five papers that seem particularly relevant as we think about the next stage of this work. These come out of the literature we've been reviewing for the current project, and each one touches on interpretability, transferability, or multi-objective behavior—areas that have come up repeatedly in our discussions.

TOP 5 PAPERS:***1. Learning Best Paths in Quantum Networks (Wang et al., INFOCOM 2025)***

This provides BeQuP as a solid neural baseline for routing. Their results help frame what neural models accomplish on path-only selection, which gives us a useful point of comparison when we later explore whether evolved policies can offer similar performance with more interpretable structure.

2. Multipath Inter-Domain Routing (Li et al., IEEE ToN 2025)

QPSEL's multi-AS design introduces a perspective on scalability and topology generalization that complements the direction we've been discussing. It offers a natural way to think about evaluating transferability when policies are deployed on networks they weren't explicitly trained on.

3. EXA-GP / Evolutionary Graph-Based GP (Desell et al.)

This is the methodological anchor for the evolutionary component. Something that stands out to me is its emphasis on producing explicit, inspectable decision structures, which seems like a meaningful counterpoint to the opacity of the neural approaches we've evaluated so far.

4. Best-of-Both-Worlds Bandits (Zimmert & Seldin, NeurIPS 2019)

This introduces a way of handling stochastic and adversarial behavior within the same learning framework. It's relevant as we think about whether hybrid approaches might reduce the amount of scenario-specific switching we've seen in our allocator analysis.

5. Equality of Opportunity in ML (Hardt et al., NeurIPS 2016)

Outside quantum networking, it provides the conceptual grounding for fairness as a multi-objective consideration. This could be useful if we move toward multi-objective fitness functions in later phases, especially given how central fairness is in the broader research themes we've talked about.

Taken together, these papers outline a direction where EXA-GP could eventually fit naturally alongside our existing neural bandit work—not as a replacement, but as an interpretable alternative we can evaluate on equal footing. Beginning with stochastic-only benchmarking would let us quantify variability and establish clean baselines before moving into more complex scenarios or multi-objective extensions. This feels consistent with the direction of our current paper, the email discussions, and the broader trajectory of the work so far.

At the same time, focusing on the quantum routing strategies emphasized in these papers could serve as a helpful segue—giving us immediate, testable alternatives that integrate cleanly with the current framework and potentially highlight opportunities or limitations that prepare us better for any next steps involving evolutionary approaches.

Let me know if there are any papers you'd like me to review more deeply. I'll be doing a lot of reading this week!

Thanks,
Piter